

Classifying Knot Diagrams with a Spatial–Behavioural Logic

A logic obtained for free from the encoding of knots as processes

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Abstract

A companion note encoded knot diagrams as cost-accounted rholang processes and showed the encoding invariant under the Reidemeister moves up to weak bisimulation. Weak bisimulation, however, is nearly blind on a closed knot: the encoding of a closed diagram is a live τ -network with no free names, so its observable behaviour carries almost no information. This note takes up the other half. Because the host is a graph-structured lambda theory, it comes with a logic — generated from its own constructors and rewrite rules by the OSLF family of algorithms — and we get that logic for free the moment we encode knots into it. Nothing here is designed for knots. We assemble the pieces the template supplies, add two features it does not yet have (a greatest fixed point and a graded separating conjunction), and observe that in a reflective calculus the name predicates are considerably stronger than in a calculus of atomic names, because a name is a quoted process and a predicate may therefore see into its structure. We then work a stable of examples: split diagrams, nugatory crossings and reducedness, connected sums, Conway spheres, alternation, and the writhe. Two of these are worth the price of admission on their own. Reducedness is a one-line separation statement, because a nugatory crossing is exactly a crossing whose deletion disconnects the term — a property no behavioural logic can express. And the conjunction *reduced* $\wedge$ *alternating* carves out precisely the fragment on which the cost ledger of the companion note stops being a framing and becomes a knot invariant. Throughout we are candid that these are properties of *diagrams*; we argue this is the right level, since it is the level at which classical knot theory states the Tait conjectures.

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1 Introduction

1.1 What we are claiming, and what we are not

The companion note [1] encodes a knot diagram as a process. A crossing becomes a gate: the over-strand passes a signal through and produces a private latch; the under-strand joins on its incoming signal and that latch; chirality is which strand plays which role. Driving the construction from a Dowker–Thistlethwaite code makes every arc a single-producer/single-consumer channel, so a diagram is a flat parallel composition of gates plus injected current. Taken as a monoidal functor from tangles to processes modulo weak bisimulation, the encoding is invariant under the Reidemeister moves.

That result is one-directional. It says non-isotopic diagrams are never *wrongly* separated. It says nothing about whether isotopy classes are *distinguished*, and there the news is bad: closing a tangle into a knot restricts every boundary name, so a closed encoding is a live, periodic τ -network. Weak bisimulation abstracts exactly the internal steps. As an observable object a closed knot is nearly featureless, and one should not expect its weak-bisimulation class to carry knot type.

The claim of this note is that the discriminating content is not gone; it has moved. It is in the *spatial* structure of the term — how many gates there are, which arcs they share, how the network falls apart when a gate is removed, what the chirality pattern is as one walks the traversal. All of that is invisible to a modal logic and completely visible to a spatial one. The encoding did not destroy the information; weak bisimulation is simply the wrong instrument for reading it.

What we are *not* claiming is that any of this is a knot invariant on the nose. Almost every formula below is a property of a diagram, and we take Section 5 to say plainly which is which. We regard this as the honest and also the useful level: it is the level at which Tait stated his conjectures, and the level at which the classical results about alternating diagrams live.

1.2 The logic is not designed; it is generated

The reason the logic costs nothing is worth stating up front, because it is the whole architectural point and is easy to mistake for a slogan.

A graph-structured lambda theory (GSLT) is a triple (Σ, E, R) : a signature which is in general a lambda theory, a set of equations imposing a structural congruence, and an adjoined graph of rewrite rules [2]. Given such a presentation, the OSLF family of algorithms generates a logic from it: one structural connective per term former of Σ , one modality per rewrite rule of R together with a choice of redex position, and a propositional layer stratified by the categorical strength available. The generation does not know which theory it is applied to.

The consequence for us is immediate. The companion note encodes knots into the rho calculus. The rho calculus is a GSLT with a published presentation. So the logic for knot diagrams is whatever the generator emits on that presentation — we do not get to choose it, and equally we do not have to build it. What the present note contributes is not a logic but the observation that the generated one already says the things a knot theorist wants said, plus two additions to the template (§2.3) and one observation about reflection (§2.2) that make the saying practical.

Remark 1 (Why the base theory is the encoding). One could instead present the tangle category **Tang** directly as a GSLT and generate a logic from *its* constructors and moves. The companion note declines that route and so do we, for the reason given there: the discriminating material is the gate — which strand produces the latch, whether the latch is matched or counted, how gates sit beside one another under $|$ — and gates are objects on the process side. A direct presentation of the tangle category has no gates in it to talk about. It also keeps the resolution reading of [1, §9] in play, since the four-port hole is what a gate becomes when its latch is forgotten (§3.3).

1.3 Ideal logic, checkable fragment

One methodological commitment governs everything below. We distinguish the *ideal* logic — adequate for the idealised calculus, and the thing that *defines* the invariant — from *fragments*, restricted so that model checking terminates, which *decide* it. These are different artifacts with different jobs, and it is a mistake to ask one to do the other’s work.

The precedent is familiar and should reassure the topologist. The Kauffman bracket is *defined* by a sum over 2^n resolutions; nobody computes it that way, and nobody thinks the definition is impugned by the existence of a better algorithm. Likewise here: the adequate logic says what the invariant is, and a user chasing a terminating check restricts the theory, or the connectives, or both, and accepts that the restricted logic no longer resolves everything. Since the spatial and behavioural connectives are generated from the theory, restricting the theory restricts them automatically; the connectives arising from the collection semantics — infinitary joins, and hence the fixed point of §2.3.1 — can force non-termination independently and may need restricting on their own account. A logic whose model checking terminates is necessarily not adequate for the whole of a Turing-complete host.

We flag which side of this line each example falls on as we go.

2 The logic, assembled

2.1 What the template supplies

We recall only what we use. The rho calculus has term formers $\mathbf{0}$, $*n$ (drop), $n!(Q)$ (output), $\text{for}(x \leftarrow n)P$ (input), parallel composition, and $@P$ (quote); parallel composition carries an associative–commutative equational theory, and the communication rewrite is the single interaction rule. Feeding that presentation to the generator yields three layers.

Structural connectives. One type former $f^\#$ per term former f , whose inhabitants are the terms whose root former is f . We write $\text{out}(n)\varphi$ for the predicate holding of $n!(Q)$ with $Q \models \varphi$, and $\text{in}(n)\varphi$ for the analogous predicate on receipts. The interesting one is parallel composition. Because $|$ is associative–commutative, the generated connective is a genuine *separating conjunction* in the sense of spatial and separation logics [5, 8]:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi, R \models \psi.$$

Everything in §4 that a modal logic cannot say is said with this connective.

Behavioural modalities. One primitive modality $\langle K_j \rangle^{\vec{x}::\vec{A}} B$ per rewrite rule and choice of redex position, where K_j is the one-hole context determined by that position. These are *context-labelled*: the label is the context in which the step fires, not merely an action name. We return to why that matters in §3.1.

Propositional layer. Conjunction and $\top$ come from the finite-limits fragment; disjunction, implication and negation require more, and a specification supplies them or does not. We assume conjunction, disjunction, and negation throughout the ideal logic, and say so when a fragment gives one up.

Restriction. The encoding of a closed diagram is **new** $a_1, \dots, a_{2n}$ **in** $\{\dots\}$, so every arc name is bound. We need a connective that gets under the binder. Following the spatial-logic tradition we take a hidden-name quantifier:

$$P \models \text{Hx}.\varphi \iff P \equiv \text{new } x \text{ in } Q \text{ with } Q \models \varphi, x \text{ fresh.}$$

Remark 2 (Restriction hides names from observation, not from description). This is the crux of the whole note and deserves to be isolated. Restriction makes a name unobservable: no context can communicate on it, which is exactly why a closed knot’s behaviour is all τ and why weak bisimulation sees nothing. It does not make the name *undescribable*. $\text{Hx}.\varphi$ reaches under the binder and speaks about the structure so hidden. The spatial layer therefore recovers precisely what closing the diagram took away from the behavioural layer. The companion note’s pessimism about closed encodings is a fact about $\approx$, not about the term.

2.2 Names are processes

In a calculus of atomic names, a name predicate can do two things: test equality and test freshness. Caires’ spatial logic for the π -calculus [4] is of that kind, and its name apparatus — name equality, the revelation connective, the fresh-name quantifier — is exactly what atomic names permit.

The rho calculus is reflective. Names are not atoms; a name is a quoted process, with $@$ and $*$ mediating between the two sorts and $@(*n) = n$ as the only equation. A name predicate may therefore *see into the structure of the name*:

$$n \models @ \varphi \iff n = @P \text{ with } P \models \varphi.$$

This is namespace logic [6]: a full logic on names, in which one describes a *namespace* by a property rather than enumerating it. The difference from the atomic setting is not cosmetic, and it pays three times here.

Arc names carry their traversal index. The parametric builder of [1, App. A] constructs arc k of a knot as the reflected name $@[*knot, k]$: a structured name whose components are the diagram’s unforgeable token and the arc’s position in the DT traversal. A name predicate can therefore recover both. “ a is an arc of *this* diagram” and “ a is the arc at traversal position k ” are name-level statements, not data-level ones. In an atomic-name setting the index would have to be carried as a message and the logic could speak of it only by observing traffic — which, for a closed knot, is precisely what is unavailable.

Walking the traversal is a name operation. Consequently the walk that the alternation formula of §4.3 performs — from arc k to arc $k + 1$ — can be taken at the name level rather than by searching the term for the gate that consumes a_k . The same property has a spatial reading and a nominal reading; the nominal one is dramatically cheaper to check, since it replaces a search over n gates by a computation on a name.

A name may quote a tangle. Most consequentially for §4.6, a name can quote an arbitrary process, hence an arbitrary sub-tangle. “The channel c carries a tangle satisfying φ ” is then simply $c \models @\varphi$ — a statement about the name, with no communication required. This is the same mechanism by which Rholang 1.4 types a channel by the language definition it carries [2, §9], and it is the natural way to talk about what sits on the far side of a Conway sphere.

Remark 3 (Two boundaries, one mechanism). It is worth noticing that reflection is doing for names what the separating conjunction does for processes: both are ways of describing a decomposition without executing it. $@\varphi$ opens a name to reveal the process inside; $\varphi \mid \psi$ opens a process to reveal the parts. A knot diagram is a structure made entirely of such decompositions, which is why the two connectives between them cover so much of the classical vocabulary.

2.3 Two additions to the template

The generated logic as it stands lacks two things we need. Both are independent features rather than knot-specific hacks, and both should probably migrate into the rubric of [2] rather than living here; we develop them here so that this note is self-contained.

2.3.1 A greatest fixed point

Add to the grammar a propositional variable X and the former $\nu X.\varphi$, with X occurring only positively in φ . The semantics is the standard one and requires nothing exotic: a formula denotes a collection of witnesses, the collections are ordered by inclusion, φ positive in X induces a monotone operator on that order, and $\nu X.\varphi$ denotes its greatest post-fixed point, which exists by Knaster–Tarski provided the order has arbitrary joins.

Two comments on the cost. First, in the vocabulary of [3] the demand for arbitrary joins is a constraint on the *collection* component, not on the structural or behavioural ones — the fixed point is paid for by the shape of the container in which witnesses are aggregated. That is also why ν is the first casualty of a terminating fragment: infinitary collection connectives break termination independently of how thin one makes the theory fragment.

Second, and specific to us, the addition is not optional. The gate of [1] is a recursive definition that reinstates itself at the latch join, so a gate’s own behaviour is infinite and no finite modal formula pins it. The characteristic formula of §3 is necessarily a ν -formula. This is a satisfying place for the feature to enter, since it enters on the strength of an ordinary design decision about the

object language rather than as apparatus imported to make a particular example work; alternating diagrams (§4.3) then come along as a second use of something already bought.

Remark 4 (What ν buys is uniformity). It is worth being precise about the gain, because it is easy to overstate. For a *fixed* diagram with $2n$ arcs, alternation is a finite conjunction over the arcs and needs no fixed point at all. What ν provides is a *single* formula, the same for every diagram, in place of a $2n$ -fold conjunction that must be rewritten for each n . Uniformity, not raw expressive power over a fixed input. This matters for the same reason it matters in program logics: a classifier one has to regenerate per input is not a classifier.

2.3.2 Graded separation

The separating conjunction splits a term. For knots we care about *how much* of the interface the two halves share, since that number is precisely the classical tangle-decomposition hierarchy. Define, for a set $\vec{a}$ of restricted names,

$$P \models \varphi \mid_k \psi \iff P \equiv H\vec{a}.(Q \mid R) \text{ with } Q \models \varphi, R \models \psi, |\text{fn}(Q) \cap \text{fn}(R) \cap \vec{a}| = k.$$

$\mid_k$ is derived rather than primitive: it is $\mid$ under H , with the cardinality condition expressed by the name predicates of §2.2, which can say of a name that it is an arc of this diagram and can therefore count the shared ones. We treat it as a defined abbreviation throughout.

3 Characteristic formulae

3.1 The idea, imported

Caires [4] identifies the notion we need: a *characteristic formula* for a process — a single formula satisfied by exactly the processes equivalent to it. Characteristic formulae turn an equivalence question into a model-checking question, which is the move a classifier is made of.

We import the idea and not the logic, and the distinction matters. Caires’ modalities are labelled by actions. Ours are labelled by *contexts* (§2), which is a strictly finer instrument: the label records the one-hole context in which a step fires, so two steps that perform the same action in different surroundings are distinguished. For gates this is exactly what one wants, because chirality — the whole content of a crossing — is a fact about which half of the gate sits in which context. In an action-labelled logic one recovers chirality by encoding it into a name discipline and then reasoning about names; here it is read directly off the label. Add to this the stronger name predicates of §2.2, and the two logics differ in both of the ingredients a characteristic formula is built from.

3.2 The gate

Recall the gate:

```
def Gate(oi, oo, ui, uo) = {
  new l in {
    for (s <- oi)      { oo!(*s) | l!(Nil) }
    | for (s <- ui & _ <- l) { uo!(*s) | Gate(oi,oo,ui,uo) }
  }
}
```

Its characteristic formula describes a restriction of a two-way separation, with the recursive occurrence caught by the fixed point:

$$\chi_{\text{Gate}}(oi, oo, ui, uo) := \nu X. \text{Hl}.\left(\text{in}(oi)[\text{out}(oo)\top \mid \text{out}(\ell)\top] \mid \text{in}(ui, \ell)[\text{out}(uo)\top \mid X]\right).$$

Read left to right: the term is, up to structural congruence, the restriction of a name ℓ over a parallel composition of exactly two receipts; the first is a receipt on oi whose continuation separates into an output on oo and an output on ℓ ; the second is a join on ui and ℓ whose continuation separates into an output on uo and something again satisfying X . The greatest fixed point is the right one: we want the gate to be reinstated forever, not to be well-founded.

Two features of the encoding are legible in the formula and are worth naming. Chirality is the asymmetry between the two conjuncts — which one produces ℓ and which joins on it — so χ_{Gate} and the formula for the mirror crossing differ by swapping $oi \leftrightarrow ui$, $oo \leftrightarrow uo$. And the *matched* latch of [1] is visible as the placement of X : the gate reinstates inside the under-branch, so each latch belongs to one instantiation. Had the two receipts been made persistent instead — a *counting* latch — X would not appear and the formula would be smaller and weaker.

3.3 The relaxation lattice

χ_{Gate} sits at the top of a lattice of progressively weaker formulae, obtained by forgetting parts of it. This is where a classifier’s dials live, and the lattice has a pleasing bottom.

formula	forgets	what remains
χ_{Gate}	nothing	the gate up to $\approx$
$\chi_{\text{Gate}}^{-\nu}$	the reinstatement	one passage through the crossing
$\chi_{\text{Gate}}^{\pm}$	the continuations	chirality only
$\chi_{\text{Gate}}^{\circ}$	the latch ℓ	a four-port hole

The bottom row is the point. Forget the latch entirely and what is left is a predicate on four ports with no over/under content — which is exactly the resolution-context of [1, §9], the hole into which the two Kauffman smoothings are plugged. The dynamic gate and the Kauffman smoothing are therefore not material from two unrelated halves of a subject; they are the top and bottom of one relaxation lattice, and the bracket is what one computes after relaxing all the way down. We do not develop the bracket here, but we note that the axis exists and that the classifier and the state sum are graded by the same parameter.

4 A stable of examples

Throughout, D is a diagram with n crossings, $\llbracket D \rrbracket$ its encoding, and $a_1, \dots, a_{2n}$ its arcs, with $o(i)$ and $u(i)$ the passes at which crossing i is met as over- and under-strand.

4.1 Graded separation

One scheme covers four classical notions. Because the encoding is a flat parallel composition of gates, and because the arcs are the only shared names, a decomposition of the term is a decomposition of the diagram, and the number of shared arcs is the number of points in which the separating curve meets it.

shared arcs	spatial statement	classical notion
0	$\llbracket D \rrbracket \models \varphi \mid_0 \psi$	split link
0, after deleting one gate	$\llbracket D \rrbracket \models \chi_{\text{Gate}}(\dots) \mid (\varphi \mid_0 \psi)$	nugatory crossing
2	$\llbracket D \rrbracket \models \varphi \mid_2 \psi$	connected sum
4	$\llbracket D \rrbracket \models \varphi \mid_4 \psi$	Conway sphere; site of mutation

The correspondence is not an analogy. A simple closed curve in the plane meeting the diagram transversally in k points cuts the arc set into two parts sharing exactly k arcs, and conversely a k -shared decomposition of the term exhibits such a curve. The classical hierarchy — split, composite, mutable — is a hierarchy in k , and the logic is graded by k because the separating conjunction is graded by its interface.

Remark 5 (This is what a modal logic cannot say). Bisimulation cannot count crossings, and it cannot see that a network falls into two halves that do not communicate. Two diagrams whose encodings are weakly bisimilar may have different k -decompositions for every k . Separation is where the spatial layer earns its name, and it is a fortunate accident — or rather, not an accident at all — that knot theory is largely a subject about separating curves.

4.2 Reduced diagrams

The cleanest instance, and the one to check by hand. A crossing is *nugatory* if there is a simple closed curve meeting the diagram transversally at that crossing and nowhere else; a diagram is *reduced* if it has no nugatory crossing. Reducedness is the standing hypothesis of most of the classical theorems about diagrams, and it is usually explained with a picture.

In the encoding it is a separation statement. Delete gate i from $\llbracket D \rrbracket$. The traversal cycle breaks into two paths,

$$\pi_i^+ = [a_{o(i)}, \dots, a_{u(i)-1}], \quad \pi_i^- = [a_{u(i)}, \dots, a_{o(i)-1}],$$

the second being the latch gap of [1, §5]. Crossing i is nugatory exactly when no remaining gate has ports in both paths — when, that is, the rest of the term separates with no shared arcs.

Proposition 1 (Reducedness is a separation formula). *Crossing i of D is nugatory if and only if*

$$\llbracket D \rrbracket \models \text{H}\vec{a}. \left(\chi_{\text{Gate}}(a_{o(i)-1}, a_{o(i)}, a_{u(i)-1}, a_{u(i)}) \mid (\varphi \mid_0 \psi) \right)$$

for some φ, ψ , where φ and ψ have their free arcs in π_i^+ and π_i^- respectively. Hence D is reduced iff the negation of this holds for every i , a formula of size linear in n .

Proof sketch. Left to right: a curve meeting D only at crossing i separates the other $n-1$ crossings into those inside and those outside, and since the curve crosses no arc, no remaining gate has ports on both sides; take φ, ψ to be the χ_{Gate} -conjunctions of the two families. Right to left: a 0-shared decomposition of $\llbracket D \rrbracket \setminus G_i$ exhibits two families of gates with disjoint arc support, whose union is all arcs but those at crossing i ; the two paths $\pi_i^\pm$ are then separated by a curve through i alone. $\square$

We stress how little machinery this used. Non-separability is the primitive the connective was built on; reducedness is that primitive applied to the term with one gate removed. Nothing was encoded, and no picture was needed.

4.3 Alternating diagrams

D is *alternating* if, following the traversal, one passes alternately over and under. In the encoding, arc a_k is produced by pass k and consumed by pass $k + 1$, so alternation says: whenever a is the over-out port of a gate it is the under-in port of a gate, and conversely.

Write $\text{ov}(a) := \exists b, c, d. \chi_{\text{Gate}}(a, b, c, d) \mid \top$ for “ a is the over-in port of some gate”, and $\text{un}(a)$ for the analogous under-in statement. The walk alternates, and is a greatest fixed point over the pair:

$$\begin{aligned} A_O(a) &= \exists b, c, d. (\chi_{\text{Gate}}(a, b, c, d) \mid \top) \wedge A_U(b), \\ A_U(a) &= \exists b, c, d. (\chi_{\text{Gate}}(c, d, a, b) \mid \top) \wedge A_O(b), \end{aligned} \quad \text{Alt} := \nu(A_O, A_U). A_O(a_1) \vee A_U(a_1).$$

In the first clause a is the over-in port, so the pass consuming a is an over-pass and its out-arc is b ; the next pass must then be an under-pass, which is $A_U(b)$. The second clause is the mirror.

Remark 6 (The recursion is bounded by the diagram). The traversal closes after $2n$ steps, so the fixed point is exact at its $2n$ -th unfolding: $\nu^{2n} = \nu$ here, not an approximation. Alternation is therefore *finitely* checkable per diagram in spite of ν being an infinitary connective in general. This is the good case, and it is worth understanding why it is good: the recursion follows the *structure* — the arc cycle — rather than the reduction. Recursions that follow the reduction are the unbounded ones. By §2.2 the walk can be taken on names, so the check is $O(n)$ name operations rather than a search over gates.

4.4 Tait, and where the ledger becomes an invariant

Now the two previous formulae are conjoined, and the payoff is not a curiosity.

$\text{Red} \wedge \text{Alt}$ is exactly the hypothesis of the Tait conjectures. By the theorem of Kauffman, Murasugi and Thistlethwaite [10, 11, 12], a reduced alternating diagram realises the crossing number of its link. So a diagram satisfying the conjunction carries a certificate of minimality: a formula in the generated logic, checkable in time linear in n by Remark 6 and Proposition 1, implies a statement about the knot and not merely the diagram.

The sharper payoff concerns the companion note. There the cost ledger is shown to satisfy $c(\llbracket D \rrbracket) = 2w(D)$ identically, and the Perko pair is used to insist that this is a *framed* invariant: two reduced 10-crossing diagrams of one knot with different writhes and therefore different ledgers. Little’s 1900 claim that the writhe of a reduced diagram is a knot invariant is false, and the Perko pair is the counterexample.

But Little’s claim is true on the alternating fragment. By the flying theorem of Menasco and Thistlethwaite [13], any two reduced alternating diagrams of the same link are related by flypes, and a flype preserves the writhe. Hence:

Proposition 2 (The formula carves out the fragment). *On diagrams satisfying $\text{Red} \wedge \text{Alt}$, the cost ledger $c(\llbracket D \rrbracket) = 2w(D)$ is an invariant of the knot type, not merely of the framing.*

This is the demonstration we would lead with. A spatial-behavioural formula — one conjunct from the separating conjunction, one from the fixed point, neither designed for the purpose — identifies exactly the region on which a quantity from the *other* note upgrades from a framed invariant to an unframed one. The Perko pair, which is non-alternating, is outside the region, exactly as it must be. Two independently developed pieces of apparatus agree about where the boundary lies, and neither was tuned to make them agree.

4.5 The writhe as a counting formula

The writhe itself can be written as a formula, but only after one is careful about where the sign lives.

The sign ε_i of a crossing is *not* its chirality: chirality is which strand goes over, whereas the sign needs that together with both strand directions [1, Rem. 2]. So the plain gate term does not contain the sign, and no formula over $\llbracket D \rrbracket$ can count the writhe. In the cost-accounted theory $C(\text{rho})$, however, the token stack is a sort of the grammar [2, §6] and each crossing’s located stack carries ε_i . The sign is then part of the term, hence visible to the structural layer, and

$$W_w := \bigvee_{\substack{p+m=n \\ p-m=w}} \underbrace{\chi_{\text{Gate}}^+ \mid \cdots \mid \chi_{\text{Gate}}^+}_p \mid \underbrace{\chi_{\text{Gate}}^- \mid \cdots \mid \chi_{\text{Gate}}^-}_m \mid \top$$

says that the diagram has writhe w , where $\chi_{\text{Gate}}^\pm$ is the gate formula conjoined with the predicate on its located stack’s sign.

The interest is not the formula, which is routine, but the cross-check. The same quantity now has two independent readings — a ledger reading, obtained by running the process and summing debits, and a logical reading, obtained by inspecting the term without running it — and they must agree. That they do is a small piece of evidence that the encoding and the generated logic are describing the same object, and cheap demonstrations of two routes to one answer are worth more to a sceptical reader than an additional theorem.

4.6 Mutation: the open case

The $k = 4$ row of §4.1 is the interesting risk, and we record it as an open problem rather than a result.

Conway mutation cuts a diagram along a sphere meeting it in four points, rotates the tangle inside, and reglues. It is the classical source of hard-to-separate pairs: the Kinoshita–Terasaka and Conway knots are mutants, share a Jones polynomial, share an Alexander polynomial, and are distinguished only by considerably heavier machinery.

In our setting a mutation is a re-plugging of a four-arc subterm — and by §4.1 the site is *visible*: $\llbracket D \rrbracket \models \varphi \mid_4 \psi$ locates it, and by §2.2 the tangle on the far side can be named by a quotation and described by $@\varphi$ without communicating with it. Whether the induced equivalence *separates* mutants is a different question, and it is open.

Conjecture 1 (Mutants). *There is a formula in the graded-separation fragment satisfied by the Kinoshita–Terasaka diagram and not by the Conway diagram.*

We flag this as the natural first computation for the classifier, precisely because it can come out either way and one would want to know which before building further on it. If it holds, it is a place where the process invariant strictly beats the polynomial, on the same graded-separation scheme as the easy cases of §4.2; if it fails, the failure locates a definite limit and is worth as much.

5 Diagrams, not knots

Every formula above is a predicate on diagrams. Alt detects an alternating *diagram*; “ K is an alternating knot” is an existential over the diagrams of K . Red, the $\mid_k$ statements, and W_w are all likewise diagram-level. We have said this once per example and say it once more in general, because the companion note had to correct exactly this confusion in another guise — reading the

writhe, a diagram quantity, as a knot quantity — and we would rather not reintroduce it in new clothes.

We regard the level as correct rather than as a limitation, for three reasons.

First, it is where the classical theory lives. Reducedness, alternation, flypes, the crossing number of a *diagram*, the Tait conjectures — these are all diagram-level notions, and the theorems that make them valuable are theorems about diagrams that have knot-level consequences. Section 4.4 is precisely of that shape.

Second, the passage from diagram to knot is a quotient, and quotients are downstream work. A diagram predicate φ induces two knot predicates, $\exists D \in K. \varphi(D)$ and $\forall D \in K. \varphi(D)$, and which one is wanted depends on the application. The interesting mathematics is in showing that a particular φ has one of these forms computable — which for alternation is a genuine theorem and not a formality.

Third, the honest statement of what the companion note proved is also diagram-level in this sense. Reidemeister invariance says the encoding does not distinguish diagrams it should not. The classifier’s job is the converse, and the converse is naturally attacked by finding diagram properties fine enough to separate, then asking which of them descend.

6 Checkability

Following §1.3, we record what a terminating fragment would keep and give up.

Keep. The structural connectives, including $|$ and the derived $|_k$; the name predicates of §2.2; the context-labelled modalities; conjunction and disjunction; and ν *under a fixed unfolding bound*. By Remark 6 the recursions this note actually uses follow the traversal and are exact at $2n$ unfoldings, so the bound costs nothing on these examples.

Give up. The composition adjunct $\varphi \triangleright \psi$. Validity for spatial logics with the adjunct is where undecidability bites hardest, and the generated structural layer of [2] does not currently supply a right adjoint to $|$ in any case. We therefore place the adjunct in the ideal logic only and exclude it from the checkable fragment explicitly, rather than leaving its status implicit. Negation is the other candidate: it is what adequacy needs and what decidability least likes, and Red as stated in Proposition 1 uses it. Since the negation there is applied to a formula of bounded size over a fixed finite diagram, it is harmless; we note that this is a property of the use and not a general licence.

What terminates. On a fixed diagram with n crossings, Alt is $O(n)$ name operations, Red is n separation tests each over a term of n gates, and W_w is a single pass over the located stacks. The $|_4$ search of §4.6 is the expensive one, since the number of four-element arc interfaces is $\binom{2n}{4}$; whether it admits a better algorithm is open and is the practical question standing between this note and a usable tool.

7 Open problems

1. **Mutants.** Conjecture 1: does the graded-separation fragment separate the Kinoshita–Terasaka and Conway diagrams? This is the first computation to do.

2. **Descent.** Which of the diagram predicates of §4 have computable existential or universal closures over an isotopy class? Alternation is the model case and is a theorem; reducedness is the next one to ask about.
3. **Faithfulness.** The companion note reformulates the converse of its invariance theorem as the question whether the encoding *reflects* as well as preserves context-labelled transition structure. The present note’s formulae are candidate separating witnesses; assembling them into a proof, or into a counterexample, is the main open problem behind both notes.
4. **The relaxation lattice and the bracket.** Section 3.3 exhibits χ_{Gate}° — the gate with its latch forgotten — as the Kauffman resolution-context. Making the passage from χ_{Gate} to χ_{Gate}° into a statement relating the classifier and the state sum, rather than an observation that both live on one axis, is the most promising structural question here.
5. **Migration.** The fixed point of §2.3.1 and the graded separation of §2.3.2 are independent of knots and belong in the rubric of [2]. Whether $|_k$ should be primitive there or remain derived is a question about the rubric, not about this application.

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